

Problem Set 2 ANSWERS

Due on Gradescope: check the course Drive calendar
Consult the Late Policy in the syllabus

Instructions: Work with this file as a template, perhaps by File: Download: Microsoft Word. Other options, like handwriting, are fine as long as they are legible. Name and SID at the top.

Did you work with other students? Identify them below. **If you work with others, you should write answers in your own words.** (If you understand your answers, this should be easy.)

Type your name to sign the Honor Code. Save as a PDF, upload to Gradescope, and click through to locate each question part before clicking Submit. Profit.

You will see point totals associated with each question. But we will score your problem sets based on completion, not correctness. The point totals are there to make you familiar with how the exams will work, which are scored on correctness.

Questions below:

- [1. \[18 points total\] Short questions with answers so low and a little Roman.](#)
- [2. \[40 points total\] Immigration and capital remittances in the Solow Model.](#)
- [3. \[18 points total\] The Romer Model and the Zombie Apocalypse.](#)

Did you work with other students? List them below:

Please type your name as an affirmation of the Honor Code at the University of California, Berkeley.

“As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others.”

Type your name:

1. [18 points total] Short questions with answers so low and a little Roman.

- (a) [2 points] Recall that real GDP measured on the expenditure side is given by

$$Y = C + I + G + NX \quad (1.1)$$

What happens if there is no government and no foreign trade? Rewrite equation (1.1) and briefly explain what it states about the disposition of income, Y . Does a component of income become interpretable as something else? Explain.

When $G = 0$ and $NX = 0$, they drop out of the equation, leaving $Y = C + I$. Income now equals either consumption spending or investment spending, and we can interpret **investment as saving**, the other activity done with income besides consuming.

- (b) [2 points] In the U.S., what can you say about the level of the long-run share of consumption in GDP, meaning C/Y ? Briefly discuss.

The long-run share of consumption in GDP is **roughly constant** at about **70% or two-thirds** of GDP. You should also know roughly how big the share is. It is above 50% and below 75%, and two thirds is a pretty good midpoint to remember.

- (c) [2 points] Consumption goods and services include things like food and health care. Investment goods and services include container cranes for seaports, architectural services, and the office buildings themselves. Do the same workers in the economy typically produce both consumption goods and investment goods? Explain.

Broadly, **NO**, workers producing consumption goods are different than workers producing investment goods. Even when they are the same, like a construction worker with a backyard garden and chicken coop, these **workers could not produce both types of goods at the same time**. Modern techniques typically mean that a worker could not be very productive at both types of production. "Gig economy" workers could potentially engage in either sector, but it would have to be an architect or software engineer who also drives Uber at night.

- (d) [2 points] Given your previous answers, what should be true in the short run about the U.S. share of labor engaged in producing consumption goods, if consumption and investment sectors require and benefit from the same amount of capital and technology?

The share of labor in consumption goods should probably be constant in the short run under these circumstances. This question is intended to spur your thinking about this. Labor in the Romer model is split in this way and unchanging. In the U.S., I suspect the share of labor in consumption production has likely been rising, because productivity advances in investment production have probably been faster.

- (e) [2 points] Recall that when production is Cobb-Douglas with a capital share $\alpha = 1/3$, the marginal product of capital is given by:

$$MPK = \frac{1}{3} \bar{A} \left(\frac{L}{K} \right)^{2/3} \quad (1.2)$$

Suppose we knew these calibrated¹ values for A, K, and L:

A = 1,983

K = 80,823,096

L = 163

“Total Factor Productivity” is a residual

The U.S. capital stock is about \$80 trillion

U.S. employment is about 163 million

Using a calculator or a spreadsheet, insert these values into the right-hand side of equation (1.2) and report and comment on what you calculated. In words, what does your value mean? Round to the nearest thousandth = tenth of a percent and interpret.

This definitely requires a calculator or spreadsheet! Here's what I get:

$$MPK = \frac{1}{3} \bar{A} \left(\frac{L}{K} \right)^{2/3} = \frac{1}{3} \times 1983 \times \left(\frac{163}{80823096} \right)^{2/3} \approx \frac{1}{3} \times 1967 \times 0.00016 \approx 0.106$$

In words, the answer is a marginal product of capital of **about 10.6%**. This is the **real gross marginal product of risky capital before taxes**. It is similar to the long-run risk-free real interest rate that we expect to be about 3%, but the rate of depreciation $\delta \approx 0.04$ produces a net real marginal product of capital that is 4 percentage points lower, and taxes also cut into it further. As you may learn in ECON 136 or similar courses, a risk premium raises the real net marginal product of risky capital above the risk-free real rate of interest.

- (f) [2 points] Imagine that employment is reduced by 10 million workers to 153 million. Assuming nothing else changes, repeat your calculation above and report the MPK. What has happened? Why? Express the change in level and in percentage terms.

$$MPK = \frac{1}{3} \bar{A} \left(\frac{L}{K} \right)^{2/3} = \frac{1}{3} \times 1983 \times \left(\frac{153}{80823096} \right)^{2/3} \approx \frac{1}{3} \times 1967 \times 0.00015 \approx 0.101$$

The marginal product of capital falls to **10.1%**, a reduction of $10.6 - 10.1 = 0.5$ **percentage point**, which is pretty big, which is also $0.5/10.6 = 4\%$ **decrease**. The marginal product of capital has fallen because the amount of labor has fallen. Labor complements capital, so a reduction in labor makes capital less productive, and the marginal product of capital falls in response.

¹ The [Penn World Table 11.0 \(2026\)](#) reports current-price capital stock in millions of PPP adjusted 2021 US\$ (variable “cn”) as \$80,823,096 in 2023, with expenditure side GDP (“cgdp”) as \$25,586,132 million, for an aggregate K/Y = 3.16. In early 2026, BLS reported 163 million employed. Solow model calibration by Matthew Shapiro’s comment in Eric M. Engen & R. Glenn Hubbard (2005) “Federal Government Debt and Interest Rates,” [NBER Macroeconomics Annual 2004, \(19\): 83-160](#) is gratefully acknowledged.

(g) [2 points] Recall that GDP is given by:

$$Y = A K^{1/3} L^{2/3} \quad (1.3)$$

Use the calibrated values above (on the previous page) to calculate Y and Y' with A and K , and $L = 163$ and $L' = 153$. Report those levels of GDP and describe what you have found. What is the percentage change in GDP caused by this reduction in labor? Also, what was the percentage reduction in labor?

With these calibrated values, we find:

$$Y = A K^{1/3} L^{2/3} = 1983 \times 80,823,096^{1/3} \times 163^{2/3} \approx 25,583,480$$

$$Y' = A K^{1/3} L'^{2/3} = 1983 \times 80,823,096^{1/3} \times 153^{2/3} \approx 24,526,119$$

In words, GDP falls from about **$Y = \$25.6$ trillion to $Y' = \$24.5$ trillion**. This would be about a **4% reduction in GDP**. By comparison, **labor has fallen by about 6%**. In the math above, I left out the common factor of 10^6 for K and L . If you include it, your answer will also have that factor inside it.

(h) [2 points] Take the growth rate of both sides of equation (1.2) and briefly explain what you see.

$$g[MPK] = g\left[\frac{1}{3}\bar{A} K^{1/3} L^{2/3}\right] = g[1/3] + g[\bar{A}] + g[L^{2/3}] - g[K^{2/3}] = g[\bar{A}] + \frac{2}{3}g[L] - \frac{2}{3}g[K]$$

The growth rate of the marginal product of capital equals the growth rate of TFP (A) plus two thirds the growth rate of labor minus two thirds the growth rate of capital.

(i) [2 points] Are your answers to parts (f) and (g) and (h) consistent? Use your answer to (h) to predict what the percentage change in MPK and compare it to what you calculated. Explain.

Yes, the calibrated results and the theoretical results are consistent. We found a 4% reduction in MPK , and our growth rate rules showed that it should indeed have fallen by 4% when L falls by 6%, because 4 is two-thirds of 6.

2. [40 points total] Immigration and capital remittances in the Solow Model.

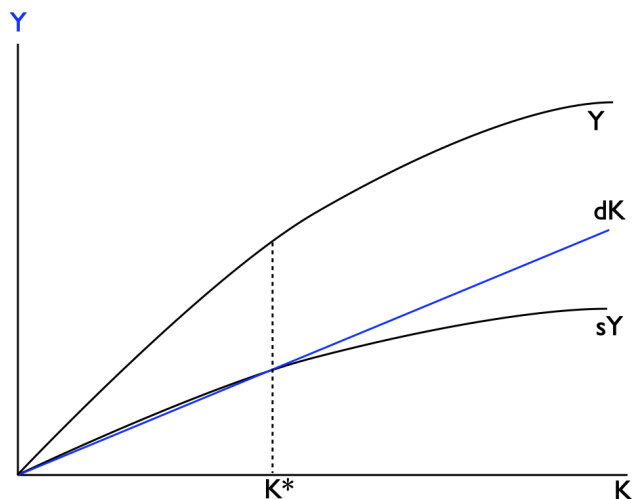
The saving rate plays a key role in the Solow Model. In particular, when the economy is closed and there are no cross-border exchanges, saving directly fuels domestic investment and increases the domestic capital stock. But in a world with mobile labor and capital, what are the implications of cross-border flows for well-being?

Imagine the U.S. is a Solow economy, and consider the effects of a one-time flow of immigration. Recall that when labor is not growing, the key equations of the Solow Model are:

$$Y = A K^{1/3} L^{2/3} \quad (2.1)$$

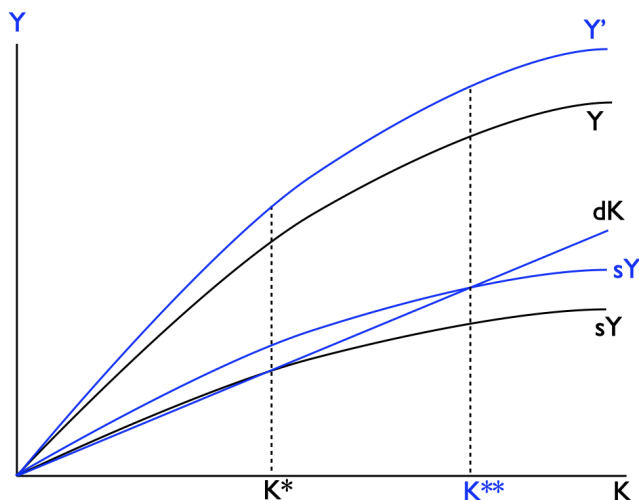
$$\Delta K = s A K^{1/3} L^{2/3} - dK \quad (2.2)$$

- (a) [4 points] On the axes below, draw the Solow diagram identifying the saving, depreciation, and income curves, and the steady state. Briefly explain what you have drawn.



This is the Solow diagram in levels of K and Y. In the textbook and in the class slides, we also see the Solow diagram in capital per worker $k = K/L$ and output per worker $y = Y/L$. In both cases, the output curve and the saving curve are concave, increasing at decreasing rates, while the depreciation line is straight with slope equal to the depreciation rate d (if the population is not growing).

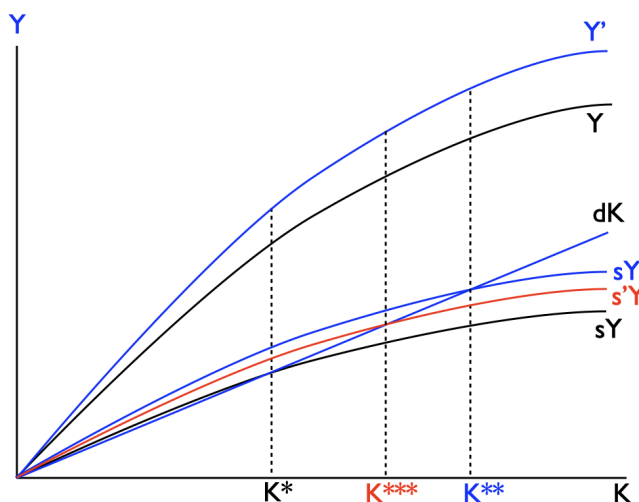
- (b) [6 points] Now redraw the Solow diagram showing the effect(s), if any, of a one-time increase in the labor supply, L , due to a substantial amount of immigration. Assume nothing else changes, and explain, discussing short and long-run effects.



When L rises, both the output curve and the saving curve shift upward because more labor results in more output given the amount of capital and productivity. In the short run, output jumps up immediately even though capital is fixed at the old steady-state level of K^* . Over time, more capital is accumulated as the economy transitions toward a higher steady state at $K^{**} > K^*$. Intuitively, the arrival of more labor raises output immediately and also even more in the long run, as laborers earn income and save a fraction, increasing the capital stock.

So far, we have assumed that immigrants save at the same rate as natives, which seems reasonable. But what would happen if immigrants sent some of their saving back to their native countries, where it became foreign capital?

- (c) [4 points] Redraw part (b) showing the full effects of immigration: the increase in L and any change in the national investment rate s owing to immigration. On the next page, explain what you have drawn, discussing short and long-run effects.



This is more art than science. I have drawn a new saving curve $s'Y'$ that is below the sY curve from part (b) because here, the saving rate has fallen. If immigrants send some of their saving back home, it does not become capital investment and must show up as a reduction in the national saving rate. That creates a third steady state K^{***} that lies somewhere to the left of K^{**} . Whether it lies between K^* and K^{**} , as I have drawn it here, or further left even below K^* as well is less clear. If the reduction in the saving rate is fully 10% as in the later part of this problem, then K^{***} will actually lie to the left of K^* because per the answer to part (g), the saving rate affects steady-state K more strongly (an exponent of $3/2$) than does L . What can we say for sure? With saving diverted abroad, the economy transitions toward a lower steady state than would have prevailed if the immigrants invested their saving domestically. It could even be a lower steady state than existed originally. But it is still the case that output jumps upward at first, in the short run.

(d) [2 points] In the steady state, what is true about ΔK ? Rewrite equation (2.2) using this.

It's equal to zero because K^* is unchanging in the steady state, so therefore $sY = dK$.

(e) [2 points] Substitute equation (2.1) into your answer to part (d) to find an equation that includes A , K , L , s , and d .

Substituting for Y gives us $s A K^{1/3} L^{2/3} = d K$

(f) [2 points] Manipulate your answer to part (e) to find an equation for the steady-state level of capital, K^* , in terms of A , L , s , and d . Show your work.

$$s A K^{1/3} L^{2/3} = d K$$

$$\frac{s A K^{1/3} L^{2/3}}{K} = \frac{d K}{K}$$

$$\frac{s A L^{2/3}}{K^{2/3}} = d$$

$$s A L^{2/3} = d K^{2/3}$$

$$\frac{s A}{d} L^{2/3} = K^{2/3}$$

$$\left(\frac{s A}{d}\right)^{3/2} L = K^*$$

Let's divide both sides by K

Now simplify

Now multiply both sides by $K^{2/3}$

Now divide by d

And raise both sides to the $3/2$ power

Done. Maybe flip: $K^* = \left(\frac{s A}{d}\right)^{3/2} L$

- (g) [2 points] Take the growth rate of both sides of your answer to part (f) to find a formula for the growth rate of K in terms of the growth rates of A, L, s, and d. Simplify.

$$\begin{aligned}
 g[K^*] &= g\left[\left(\frac{sA}{d}\right)^{3/2} L\right] \\
 g[K^*] &= g\left[\left(\frac{sA}{d}\right)^{3/2}\right] + g[L] \\
 g[K^*] &= \frac{3}{2}g\left[\frac{sA}{d}\right] + g[L] \\
 g[K^*] &= \frac{3}{2}g[s] + \frac{3}{2}g[A] - \frac{3}{2}g[d] + g[L]
 \end{aligned}$$

- (h) [2 points] Take the growth rate of both sides of equation (1) to find a formula for the growth rate of output, Y, in terms of the growth rates of A, K, and L. Simplify.

$$\begin{aligned}
 g[Y] &= g[A] + g[K^{1/3}] + g[L^{2/3}] \\
 g[Y] &= g[A] + \frac{1}{3}g[K] + \frac{2}{3}g[L]
 \end{aligned}$$

- (i) [2 points] (i) (2 points) Finally, subtract the growth rate of L from your answer to part (h) to find a formula for the growth rate of output per person, Y/L, in terms of the growth rates of A, K, and L. Simplify.

$$\begin{aligned}
 g[Y/L] &= g[Y] - g[L] \\
 g[Y/L] &= g[A] + \frac{1}{3}g[K] + \frac{2}{3}g[L] - g[L] \\
 g[Y/L] &= g[A] + \frac{1}{3}g[K] - \frac{1}{3}g[L]
 \end{aligned}$$

Today, the [share of Americans who are foreign-born is around 15-16 percent](#). For simplicity, let us assume that statistic is 10 percent, and although immigrants actually arrive in much smaller numbers, let us examine the effects of a one-time increase in L equal to 10 percent.

- (j) [3 points] Ignoring any change in the national saving rate, what happens to output per person in the short run when L rises by 10 percent? Use your answer to part (i) and be specific.

Our answer to part (i) indicates that in the short run, when capital is fixed, a 10 percent increase in labor will reduce output per person by a third of that, or by 3.3 percent.

- (k) [3 points] Ignoring any change in the national saving rate, what happens to output per person in the long run when L rises by 10 percent? Use your answers to parts (g) and (i) and be specific.

Our answer to part (g) shows that capital ultimately rises one-for-one with labor in the long run. If capital rises at the same rate as labor, then by our answer to part (i), there is no long-run effect on output per person.

Now let us think about potential effects of immigration on saving and capital. Assume the original saving rate s among U.S. natives was 20 percent, and let us take an extreme view: that all immigrants send all of their saving abroad. Thus their contribution to the national investment rate is zero.

- (l) [3 points] If 90 percent of residents save at 20 percent, and 10 percent of residents save at 0 percent, what is the new national or average saving rate, or is it the same? (Hint: this is like reducing the number of savers by the percent who are immigrants.)

Using the hint, this is a 10 percent reduction in the savers and in their saving rate, from 20 percent to 18 percent because 10 percent of 20 is 2. You could also view this as a weighted average: $0.9 \times 0.2 + 0.1 \times 0 = 0.9 \times 0.2 = 0.18$.

- (m) [5 points] What are the long-run effects of immigration on output per person if the saving rate does as you have described in part (l)? Use your answers to parts (g) and (i) and be specific.

If the saving rate also falls by 10 percent, then we see from the answer to part (g) that it contributes a reduction in capital of a size equal to $3/2$ that rate, or 15 percent! Part of this is offset by the 10 percent increase in capital due to 10 percent more labor, for a net reduction in capital of 5 percent. Looking at our answer to part (i), this indicates that output per person must have fallen by a total of 5 percent, which is the sum of a 1.7 percent (five thirds) reduction due to a 5 percent fall in capital and a 3.3 percent reduction due to a 10 percent rise in labor.

3. [10 points total] The Romer Model and the Zombie Apocalypse.

Braaaaaaaains! Consider an “old school” Romer economy² like the one we saw in class, with production of knowledge A and food Y . Scientists produce new knowledge using old knowledge and have productivity $\bar{z}$. Total population is fixed at $\bar{N}$ and employed either in knowledge or in food. The share of population in knowledge production equals ℓ , a number like 0.10.

$$\Delta A_t = \bar{z} A_t L_{at} \quad (3.1)$$

$$Y_t = A_t L_{yt} \quad (3.2)$$

$$\bar{N} = L_{at} + L_{yt} \quad (3.3)$$

$$L_{at} = \ell \bar{N} \quad (3.4)$$

Recall that the solution to the Romer Model reveals the following:

$$g[A] = \bar{z} \ell \bar{N} \quad (3.5)$$

$$A_t = A_0 (1 + g[A])^t \quad (3.6)$$

$$y_t = A_0 (1 + g[A])^t (1 - \ell) \quad (3.7)$$

Imagine what might happen if all the scientists accidentally infect themselves with the “zombie flu” and become zombies. Zombies:

- Do not produce new ideas
- Do not destroy old ideas
- Eat brains and do not eat food produced by farmers
- Do not eat farmers’ brains. They prefer tasty imported brains

(a) [3 points] When the scientists become zombies, what happens to the growth rate of A ? What about the growth rate of y ?

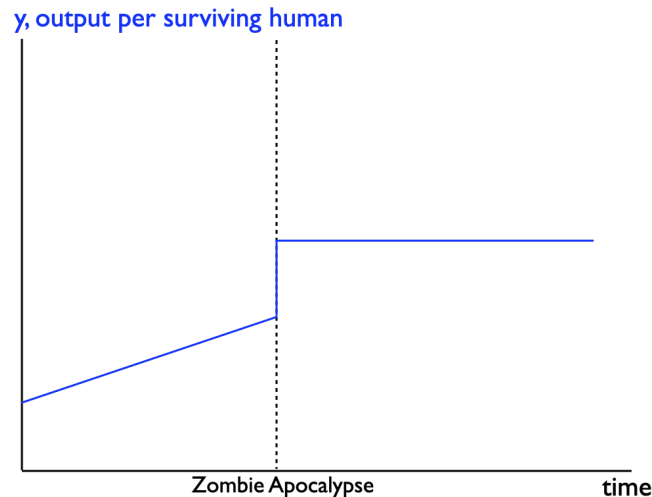
They both fall to zero because the number of scientists engaged in idea production equals 0. The growth rate of A is given in equation (3.5)

(b) [3 points] When the scientists become zombies, what happens to the level of A ? What about the level of output per person? (Consult the bullet points about zombies above.)

The level of A does not change, because zombies do not destroy knowledge. The level of output per person y actually rises because output is a function of knowledge A and food employment L_{yt} . Neither of those two inputs changes, but when all scientists L_{at} become zombies, they cease being people and part of N . So while output remains unchanged, the number of people (or “food mouths”) falls, which raises y . Losing scientists to zombiedom is bad for idea production but actually good for living standards, because there is more food per farmer.

² The Romer model in Jones 6e is a little different. The “old school” Romer model is what appears in editions 1-5 and in the course slides for ECON 100B.

- (c) [4 points] Illustrate the Zombie Apocalypse in the graph below, showing any changes in the levels and growth rates in output per person over time. Discuss what you have drawn. Is the zombie flu good or bad? Explain.



Here, there is an interesting difference between the level and growth effects. Scientists haven't been repurposed, which would have a different effect on output per person. Instead, they just go away. The dependency burden on the farmers, who previously had to give some of their farm output to scientists, is reduced. Output per person rises. But no more scientists means no more new ideas. Technology stagnates, and so will output per surviving person. Hence the picture, which shows an upward jump in production per surviving person, and then a plateau. The Zombie Apocalypse is good in the short run, as we shed scientists and have more to eat ourselves (pun intended again). But in the long run, no scientists means no increase in production per survivor, which is a glum tale indeed.

- (d) [4 points] Suppose rather than being turned into zombies, the scientists had all been deported to another country by the government. Would anything be different? Why?

Deportation of scientists would look the same as the Zombie Apocalypse, because both result in having no scientists to feed and also no scientists to develop new ideas. (If it were costly to deport the scientists, that might reduce y temporarily, as farmers would have to stop farming and deport the scientists.)

- (e) [4 points] Suppose rather than being turned into zombies, the scientists had all been repurposed as farmers by the government. Would anything be different? Why?

Repurposing scientists into farming would increase farm output by $\frac{1}{2}$, but those scientists would also need to eat food. Population does not change, whereas before it fell by $\frac{1}{2}$. So output per person actually does *exactly the same thing it did before*. Output Y now jumps higher when the scientists are repurposed, while it didn't do that before, but population also does not jump downward. The net effect on y is a wash.